Zoo202 final term

Subjective fall 2022

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The cochlea

Is a hollow, spiral-shaped bone found in the inner ear that plays a key role in the sense of hearing and participates in the process of auditory transduction. Sound waves are transduced into electrical impulses that the brain can interpret as individual frequencies of sound.

The pelvic girdle three name.

Also known as the hip bone, is composed of three fused bones: the ilium, ischium and the pubic bone. The pelvic spine is the posterior portion of the pelvis below the lumbar spine, composed of the sacrum and coccyx.

Nonshivering thermogenesis

Is defined as an increase in metabolic heat production (above the basal metabolism) that is not associated with muscle activity. It refers mainly to the increased metabolism of brown fat, but to a lesser degree can also be detected in skeletal muscles, liver, brain, and white fat.

Ancient bird found in China.

Gansus yumenensis is the first Mesozoic bird found in China. It was identified following the discovery of a fossil near Changma, Gansu Province, northwestern China in 1984.

Alligator and crocodile

The main thing to remember is that alligators and crocodiles are different species. They can be identified by their jaw shape, size and color. Alligators are dark green or black, have u-shaped snouts, tiny black spots and are smaller than crocs. Crocodiles are bigger, have v-shaped snouts and are more aggressive

e difference between proventriculus and ventriculus.

Is that proventriculus is the part of the avian stomach, between the crop and the gizzard, that secretes digestive enzymes while ventriculus is a digestive cavity or stomach.

Negative pressure mechanism

Most reptiles breathe by changing the volume of the body cavity. By contractions of the muscles moving the ribs, the volume of the body cavity is increased, creating a negative pressure, which is restored to atmospheric level by air rushing into the lungs.

Characteristics of Mammals

The presence of hair or fur.

Sweat glands.

Glands specialized to produce milk, known as mammary glands.

Three middle ear bones.

A neocortex region in the brain, which specializes in seeing and hearing.

Specialized teeth.

A four-chambered heart.

Nervous system of enteropneust.

The nervous system of enteropneustsis ectodermal in originLies at the base of the ciliated epidermis. Consists of dorsal and ventral nerve tracts And a network of epidermal nerve cells, called a nerve plexus.

There are no major ganglia.

SensoryThe group of eggs laid and chicks produced by a female is called a

Clutch.

The group of eggs laid and chicks produced by a female is called a clutch

Pheromones.

Pheromones are used to recognize members of The same species, members of the opposite sex, And the reproductive state of a member of the Opposite sex

• Pheromones may also induce sexual behavior,

Help establish and recognize territories, and ward Off predators

Hibernacula.

Individuals that are usually solitary may migrate to a common site to spend the

Winter. Heat loss from these groups, called hibernacula,

Pygostyle.

The posterior caudal vertebrae are fused into a pygostyle, which helps

Support the tail feathers that are important in steering.

Ram ventilation

constitutes swimming in relatively straight lines at a high rate of speed with the mouth open. This forces water through the mouth and over the gills, meaning the fish can breathe passively while swimming. Some classic examples are tuna, swordfish and shark

Homodont –

Teeth are all about the same shape (most vertebrates, few mammals).

Heterodont –

Teeth have different form and functions in different parts of the tooth row (mammals, a few fish)

Convergent evolution

• The evolution of similar structures in different

Lineages is called convergent evolution

embryonic diapause

• Development is arrested for the three months During which the recently born young do most Of their nursing. This mechanism is called Embryonic diapause

Tunicate opening.

The adult tunicate body is shaped like a small barrel with two

Large openings called siphons.

• In mammals, the secondary palate

extends posteriorly by a Fold of skin, called the soft palate, which almost completely Separates the nasal passages from the mouth cavity.

• The lateral-line system

consists of sensory pits in the epidermis In these Pits are receptors that are stimulated by water

• Lateral lines are used to detect either water currents, or apredator

• Fishes may also detect low-frequency sounds with these receptors.

• Countercurrent heat-exchange systems

may help Regulate heat loss from exposed areas (figure 22.14)

• Arteries passing peripherally through the core of an Appendage are surrounded by veins that carry blood Back toward the body

• When blood returns to the body through these veins,

Heat transfers from arterial blood to venous blood and Returns to the body rather than being lost to the Environment

Types of flight.

Different birds, or the same bird at different times, use different kinds of flight

• During gliding flight

, the wing is stationary, and a bird loses altitude. Waterfowl coming in for a Landing use gliding flight

- **Flapping flight**

generates the power for flight and is the most common type of flying. Many Variations in wing shape and flapping patterns result in species-specific speed and Maneuverability

- **Soaring flight**

allows some birds to remain airborne with little energy expenditure. During Soaring, wings are essentially stationary, and the bird utilizes updrafts and air currents to gain Altitude.

Hummingbirds perform **hovering flight**

The adult tunicate opening

Body is shaped like a small barrel with twoLarge openings called siphons.

Secondary palate

A plate of bone, called the secondary palate, evolved in the archosaurs and separates The nasal and mouth passage ways.

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. Tunicates

Derive their name from a thick Extracellular body covering, called a tunic, that Is made up of cellulose-like, complex sugars.

Another interesting feature of tunicates is Their circulatory system.

A pineal body

On the roof of the forebrain appears to stimulate ovarian Development and to regulate other functions influenced by light and dark periods.

Urochordates example

Tunicates or sea squirts